A Major Project Report
On

"AUTONOMOUS PRECISION FARMING ROBOT FOR SMART CROP MANAGEMENT AND ANALYSIS"

Submitted in Partial Fulfillment of the Academic Requirement for the Award of Degree
of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS & COMMUNICATION ENGINEERING

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EXPLORE TO INVENT

Department of Electronics and Communication Engineering

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2025-2026

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CERTIFICATE

This is to certify that a Major Project entitled with **"Autonomous Precision Farming Robot for Smart Crop Management and Analysis"** is being submitted by:

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To JNTUH, Hyderabad, in partial fulfillment of the requirement for award of the degree of B.Tech in ECE and is a record of a bona-fide work carried out under our guidance and supervision. The results in this project have been verified and are found to be satisfactory. The results embodied in this work have not been submitted to have any other University for award of any other degree or diploma.

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EXTERNAL EXAMINER

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DECLARATION

We hereby declare that the project report entitled **“Autonomous Precision Farming Robot for Smart Crop Management and Analysis”** submitted in partial fulfilment of the requirements for the award of the degree is a record of original work carried out by us. This work has not been submitted previously, in whole or in part, to this or any other university or institute for the award of any degree, diploma, or other academic qualification. The matter embodied in this report is the result of my own efforts, except where due acknowledgment has been made in the text.

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ABSTRACT

This project presents an **Arduino-based autonomous robot** developed to support precision farming through smart crop management and analysis. The system is designed to reduce manual labor, enhance productivity, and improve crop yield by automating key agricultural operations. The robot is equipped with **ultrasonic sensors**, **DC motors**, **servo motors**, and a **16x2 LCD display**, all controlled by an **Arduino Uno microcontroller**. These components enable the robot to navigate through fields autonomously, detect and avoid obstacles, and maintain accurate crop row alignment.

In addition to navigation, the robot performs **environmental monitoring** by collecting data on parameters such as soil moisture, temperature, and humidity. The real-time data is processed by the Arduino for intelligent decision-making, enabling adaptive responses to varying field conditions. The system's **modular design** allows easy customization and scalability, making it suitable for different crop types and field sizes.

By integrating **automation, sensing, and control**, the proposed robot provides a **cost- effective and energy-efficient solution** for modern agriculture. It supports **data-driven farming practices**, optimizes resource utilization, and contributes to sustainable agricultural development through intelligent field management and analysis.

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

In recent years, agriculture has increasingly become a battleground for innovation. With growing global population, diminishing arable land per person, rising labour costs, and the unpredictable impacts of climate change, the traditional methods of crop cultivation are under ever-greater pressure. In this context, the concept of precision agriculture has emerged as a promising path: by tailoring cultivation practices to the exact conditions of a given crop, field or micro-zone, farmers can optimise input use (water, fertiliser, pesticides), reduce waste, and increase yields. Such approaches are being enabled by advances in sensing, automation and data-driven decision making.At the heart of this transformation is the deployment of autonomous field robots machines capable of navigating crop fields, detecting and responding to environmental cues, and performing tasks that would otherwise require intensive manpower. For example, robotic solutions in precision farming promise fewer chemical treatments, lower soil compaction, and more consistent coverage benefits that conventional tractors and manual labour often struggle to deliver.

In this project, we propose and develop an Arduino-based autonomous robot designed specifically for the domain of smart crop management and analysis. By equipping the system with ultrasonic sensors for obstacle detection, DC motors and servo motors for actuation.

The motivation for such a system lies in several inter-locking challenges currently facing the farming sector.First, many agricultural regions face labour shortages or excessive labour costs; automating routine tasks can reduce dependency on manual workforce, freeing human labour for more skilled tasks. Second, tasks like row-following, obstacle avoidance, and environmental monitoring are highly repetitive and can benefit from automation both in terms of consistency and precision. Third, given increasing attention to sustainable farming, technologies that reduce wasted inputs (e.g., water, fertiliser, pesticide) and minimise soil disturbance are increasingly valuable; autonomous robots enable more targeted operations, leading to lower environmental footprint. Fourth, the modularity of the design allows scaling: instead of monolithic single-purpose machines, our robot can be adapted and expanded for varying crop types,field sizes and terrains which improves flexibility and cost-effectiveness.

The design choices made in this project reflect these motivations. The use of a microcontroller platform (Arduino) offers accessibility and low cost, making the system more feasible for smaller farms or regions with limited budgets. Ultrasonic sensors provide a reliable and cost-effective means of obstacle detection, enabling safe navigation in dynamic farm environments. Motor drivers, DC motors and servo motors together deliver the necessary mobility and actuation for both traversal and manipulation of field tasks. The incorporation of a 16×2 LCD interface allows real-time display of system status, sensor readings and navigation modes an important factor in field-deployable systems where users may wish to monitor or intervene manually. Navigation strategies which include crop-row alignment allow the robot to trave l consistently between planted rows, ensuring task accuracy and minimizing damage to the crops.

Moreover, the system's modular architecture means that as needs evolve for example, to integrate additional sensors (soil moisture, temperature/humidity, multispectral vision) or additional actuators (sprayers, seeders,weeding tools) the robot can be extended. This adaptability is particularly relevant given the heterogeneity of farmland terrains, crop types, and farming practices across regions. Whether deployed in flat cereal fields, sloped orchards, or smaller vegetable plots, the robot can be tuned or scaled to provide effective assistance.With this backdrop, the remainder of this paper is organised as follows: first we review relevant literature in precision agriculture and agricultural robotics; then we describe the mechanical, electrical and software architecture of the proposed robot; next we explain the navigation, sensing and decision-making algorithms; finally we present experiments carried out in representative field setting.

In summary, this work aims to bring together accessible hardware, careful design and modular architecture to deliver an autonomous farming robot capable of performing core tasks in smart crop management. By doing so, we hope to contribute to the broader goal of enabling precision agriculture: improving efficiency, reducing resource consumption, and supporting sustainable farming practices in a world of growing food demands.

CHAPTER 2

LITERATURE SURVEY

Artificial Intelligence-Driven Autonomous Robot for Precision

In agriculture (in the context of this paper, the terms “agriculture” and “farming” refer to only the farming of crops and exclude the farming of animals), smart farming and automated agricultural technology have emerged as promising methodologies for increasing the crop productivity without sacrificing produce quality. The emergence of various robotics technologies has facilitated the application of these techniques in agricultural processes. However, incorporating this technology in farms has proven to be challenging because of the large variations in shape, size, rate and type of growth, type of produce, and environmental requirements for different types of crops. Agricultural processes are chains of systematic, repetitive, and time-dependent tasks.

However,some agricultural processes differ based on the type of farming, namely permanent crop farming and arable farming. Permanent crop farming includes permanent crops or woody plants such as orchards and vineyards whereas arable farming includes temporary crops such as wheat and rice. Major operations in open arable farming include tilling, soil analysis, seeding, transplanting, crop scouting, pest control, weed removal and harvesting and robots can assist in performing all of these tasks. Each specific operation requires axillary devices and sensors with specific functions. This article reviews the latest advances in the application of mobile robots in these agricultural operations for open arable farming and provide an overview of the systems and techniques that are used. This article also discusses various challenges for future improvements in using reliable mobile robots for arable farming.

Autonomous robots and solar energy for precision agriculture and smart farming

The growing population, demand for healthy diets, and shift towards plant-based protein diets are increasing pressure on food production and land use. However, current agricultural practices jeopardize soil health and biodiversity, which threatens future ecosystems and food production. Precision Agriculture (PA) offers a solution to these challenges by minimizing resource use and maximizing yield through a multidisciplinary approach that includes engineering, AI, and robotics. Robotics, in particular, will be critical to delivering PA and enabling sustainable, healthy food production. This book chapter discusses robotic technologies available for precision agriculture. For precision agriculture to be truly sustainable,agriculture machinery and devices used for data collection must operate on clean energy. Among renewable energy sources, solar energy and solar PV show great potential to dominate the future of sustainable agriculture development. Solar-powered devices are inevitable for developing PV in rural and off-grid agriculture farms and lands. To transition to photovoltaic agriculture, significant changes to agricultural practices and the adoption of smart technologies like IoT, robotics, and WSN are necessary. As future food production adapts to changing consumer behaviour and environmental factors, PA will play a critical role in improving sustainability by minimizing the use of diminishing resources and GHG emissions through the use of renewable energy sources. However, the adoption of new technologies should also incorporate green energy sources like solar energy to meet power requirements for the sustainable development of these smart technologies. With the influx of robotic technologies into the agriculture sector, increasing power demand is inevitable, especially in remote areas where PV-based systems can play a game-changing role. The agricultural sector is expected to witness a technological revolution towards sustainable food production, which cannot be achieved without solar PV development and support.

Autonomous robot navigation for precision horticulture

Certain experimental horticultural operations require an autonomous vehicle capable of navigating through a field of crop plants. Since the vehicle must be able to operate in many fields to be economic, artificial navigation beacons and detailed prior maps are disadvantageous. A novel navigation scheme has been devised, which allows the crop rows themselves to be used as a navigation aid. The commanded path of the vehicle through the field is expressed by path curvature as a function of forward distance; parts of the path are marked as being aligned with crop rows. Data from image analysis is combined with that from a solid state compass and dead reckoning using an extended Kalman filter (EKF).. The method has been implemented on a small horticultural vehicle, allowing it to follow crop rows and turn at the end of the rows fully autonomously.

A PV-Powered Microcontroller-Based Agricultural Robot Utilizing GSM Technology for Crop Harvesting and Plant Watering.

A solar-powered, intelligent grass cutter and pesticide application system that uses microcontrollers, sonar sensors, solar panels, and GSM technology is being proposed as a solution to the issue of high labor costs in the agricultural sector of Bangladesh. This is done in order to achieve exquisite cost management. The initial objective is to design a system that, through utilizing a robot controlled by a micro controller, will make the laborious work of our farmers and entrepreneurs more feasible. This machine can trim the grass, apply pesticides, and simultaneously provide the plant with water. Solar panels are used to extract power from the sun to charge the device. At long last, GSM technology has been integrated, turning this device into one that can be remotely controlled using a straightforward SMS protocol.

The smart image recognition mechanism for crop harvesting system in intelligent agriculture.

study proposed a harvesting system based on the Internet of Things technology and smart image recognition. Farming decisions require extensive experience; with the proposed system, crop maturity can be determined through object detection and mature crops can then be harvested using robotic arms. Keras was used to construct a multilayer perceptron machine learning model and to predict multiaxial robotic arm movements and position. Following the execution of object detection on images, the pixel coordinates of the central point of the target crop in the image were used as neural network input, whereas the robotic arms were regarded as the output side. A MobileNet version 2 convolutional neural network was then used as the image feature extraction model, which was combined with a single shot multibox detector model as the posterior layer to form an object detection model. The model then performed crop detection by collecting and tagging images. Empirical evidence shows that the proposed model training had a mean average precision (mAP) of 84%,

Iot based agriculture using agribot

The advancement in the technology reduces the farmer efforts in the agriculture fields. India is the country more than 70% of the people depends on agriculture. So, the farming in India should be innovative to moderate the hard work of the agriculturalists. Numerous amount of procedures are accomplished in the farming field like sowing, shoeing, weed cutting, cultivating etc. Very basic and important maneuver is sowing, ploughing, weed removal. But the current approaches of sowing, ploughing and plant cutting are challenging. The tools used for seed sowing are problematic and troublesome has it requires more man power. Hence, there is a necessity to advance the equipment which will decrease the efforts of the agriculturalists. The proposed system using an Agribot goals to seed at specific locations with definite space between two seeds and positions while seeding.The robotic arm helps in unwanted plant elimination. The heart of the proposed system is microcontroller which controls the entire operation. The prototype model has been implemented so that it can be scaled up for development of the larger systems.

Visual monitoring in a simulated agricultural machinery operation

Many agricultural machinery operations involve the monitoring of implements located behind the tractor operator. This visual monitoring task of a tractor operator was simulated in the laboratory. The effects of the type of task (steering only vs. steering plus rear monitoring), the operators' peripheral field of view, and the steering input frequency on operator performance were studied. Response times to visual stimuli located in the visual sphere surrounding the operator were determined. Visual field contours derived from the experimental data were plotted and used to illustrate variations in the operators field of view due to changes in the experimental variables.

A robot platform for unmanned weeding ina paddy field using sensor fusion

A robot platform which can do weeding while traveling between rice seedlings stably against irregular land surface of a paddy field. Also, an autonomous navigation technique that can track on stable state without any damage of the seedlings in the working area is proposed. Detection of the rice seedlings and avoidance knocking down by the robot platform is achieved by the sensor fusion of a laser range finder (LRF) and an inertial measurement unit (IMU). These sensors are also used to control navigating direction of the robot to keep going along the column of rice seedling consistently. Deviation of the robot direction from the rice column that is sensed by the LRF is fed back to a proportional and derivative controller to obtain stable adjustment of navigating direction and get proper returning speed of the robot to the rice column.

CHAPTER 3

IMPLEMENTATION

3.1. EXISTING METHOD:

Traditional farming relies heavily on manual labor for tasks such as crop monitoring, row alignment, and obstacle avoidance. These methods are time-consuming, prone to human error, and inefficient in large-scale operations. While some mechanized systems exist, they often lack autonomy and require constant human supervision. Moreover,existing systems rarely incorporate real-time environmental sensing or adaptive control mechanisms, leading to suboptimal resource utilization and inconsistent crop health monitoring.

3.2. PROPOSED METHOD:

The proposed system introduces a smart, sensor-driven pump controller with scheduled automation and voice-based alerts. The RTC module ensures the pump operates at predefined times. The DHT11 sensor provides environmental data to optimize pump usage. The APR33A3 module plays audio messages like “Pump ON,” “Pump OFF,” or “High Temperature Detected.” The LCD displays temperature, humidity, and pump status. This system enhances efficiency, reduces manual effort, and improves user awareness through audio and visual feedback.

BLOCK DIAGRAM

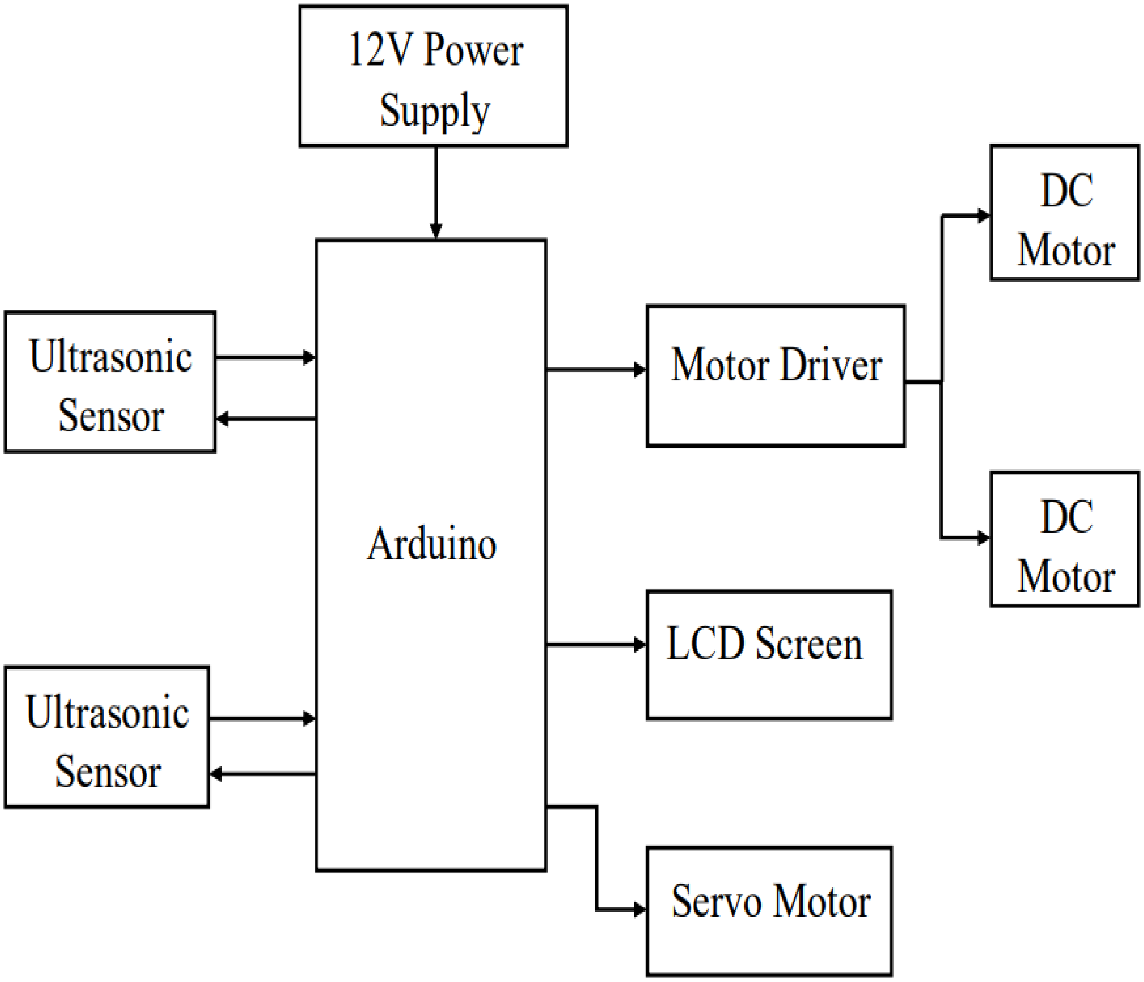

Fig 3.2.1 :Block Diagram

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CHAPTER 4

HARDWARE CONSTRAINTS

4.1. ARDUINO UNO

Fig 4.1.1 : Arduino Uno

The Arduino Uno is a series of open-source microcontroller board based on a diverse range of microcontrollers (MCU). It was initially developed and released by the Arduino company in 2010. The microcontroller board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery. It has the same microcontroller as the Arduino Nano board, and the same headers as the Leonardo board. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available.

Arduino Uno Pin description:

Pin Category	Pin Name	Details
Power	Vin, 3.3V, 5V, GND	Vin: Input voltage to Arduino when using an external power source. 5V: Regulated power supply used to power microcontroller and other components on the board. 3.3V: 3.3V supply generated by on-board voltage regulator. Maximum current draw is 50mA. GND: ground pins.
Reset	Reset	Resets the microcontroller.
Analog Pins	A0 – A5	Used to provide analog input in the range of 0-5V
Input/Output Pins	Digital Pins 0 - 13	Can be used as input or output pins.
Serial	0(Rx), 1(Tx)	Used to receive and transmit TTL serial data.
External Interrupts	2, 3	To trigger an interrupt.
PWM	3, 5, 6, 9, 11	Provides 8-bit PWM output.
SPI	10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK)	Used for SPI communication.
Inbuilt LED	13	To turn on the inbuilt LED.
TWI	A4 (SDA), A5 (SCA)	Used for TWI communication.
AREF	AREF	To provide reference voltage for input voltage.

Arduino Uno Technical Specifications:

Microcontroller	ATmega328P – 8 bit AVR family microcontroller
Operating Voltage	5V
Recommended Input Voltage	7-12V
Input Voltage Limits	6-20V
Analog Input Pins	6 (A0 – A5)
Digital I/O Pins	14 (Out of which 6 provide PWM output)
DC Current on I/O Pins	40 mA
DC Current on 3.3V Pin	50 mA
Flash Memory	32 KB (0.5 KB is used for Bootloader)
SRAM	2 KB
EEPROM	1 KB
Frequency (Clock Speed)	16 MHz

Arduino Uno is a microcontroller board based on 8-bit ATmega328P microcontroller. Along with ATmega328P, it consists other components such as rystal oscillator, serial communication, voltage regulator, etc. to support the microcontroller. Arduino Uno has 14 digital input/output pins (out of which 6 can be used as PWM outputs), 6 analog input pins, a USB connection, A Power barrel jack, an ICSP header and a reset button.

How to use Arduino Board

The 14 digital input/output pins can be used as input or output pins by using pinMode(), digitalRead() and digitalWrite() functions in arduino programming. Each pin operate at 5V and can provide or receive a maximum of 40mA current, Out of these 14 pins, some pins have specific functions as listed below:

- **Serial Pins 0 (Rx) and 1 (Tx):** Rx and Tx pins are used to receive and transmit TTL serial data. They are connected with the corresponding ATmega328P USB to TTL serial chip.
- **External Interrupt Pins 2 and 3:** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- **PWM Pins 3, 5, 6, 9 and 11:** These pins provide an 8-bit PWM output by using analogWrite() function.
- **SPI Pins 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK):** These pins are used for SPI communication.
- **In-built LED Pin 13:** This pin is connected with an built-in LED, when pin 13 is HIGH – LED is on and when pin 13 is LOW, its off.
    - Along with 14 Digital pins, there are 6 analog input pins, each of which provide 10 bits of resolution, i.e. 1024 different values. They measure from 0 to 5 volts but this limit can be increased by using AREF pin with analog Reference() function.
- Analog pin 4 (SDA) and pin 5 (SCA) also used for TWI communication using Wire library.
    - Arduino Uno has a couple of other pins as explained below:
- **AREF:** Used to provide reference voltage for analog inputs with analog Reference() function.
- **Reset Pin:** Making this pin LOW, resets the microcontroller.

Arduino Uno to ATmega328 Pin Mapping

When ATmega328 chip is used in place of Arduino Uno, or vice versa, the image below shows the pin mapping between the two.

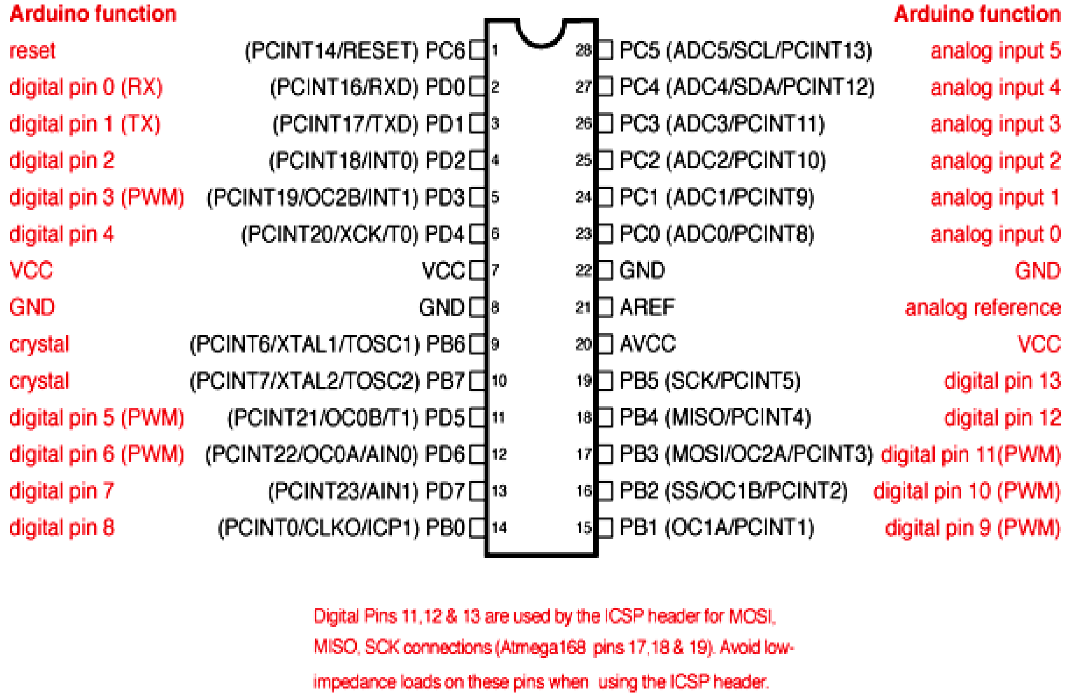


Fig 4.1.2: Arduino Uno to ATmega328 Pin Mapping

Software

Arduino IDE (Integrated Development Environment) is required to programthe Arduino Uno board.

Programming Arduino

Once arduino IDE is installed on the computer, connect the board with computer using USB cable. Now open the arduino IDE and choose the correct board by selecting Tools>Boards>Arduino/Genuino Uno, and choose the correct Port by selecting Tools>Port. To get it started with Arduino Uno board and blink the built-in LED, load the example code by selecting Files>Examples>Basics>Blink. Once the upload is finished, Below is the example code for blinkin.

Applications

- Prototyping of Electronics Products and Systems
- Multiple DIY Arduino Projects.
- Easyto use for beginner level DIYers and makers.
- Projects requiring Multiple I/O interfaces and communications.

4.2 Ultrasonic Sensor (HC-SR04)

The Ultrasonic Sensor works on the principle of reflection, let us see how

- The sensor is used to emit waves known as ultrasound with a frequency of 40 KHz.
- The transmitted wave moves in a straight line path and gets obstructed byan object if present.
- fter being obstructed, the wave reflects and is detected by the sensor. The time of transmission and reflection of the wave is noticed and the distance of the object is calculated using the speed- time formula.

Fig 4.2.1: Ultrasonic Sensor (HC-SR04)

Ultrasonic sensor Pinout:

Ultra sonic sensor is a 4 pin sensor, theyare VCC, Trigger, Echo, Ground.

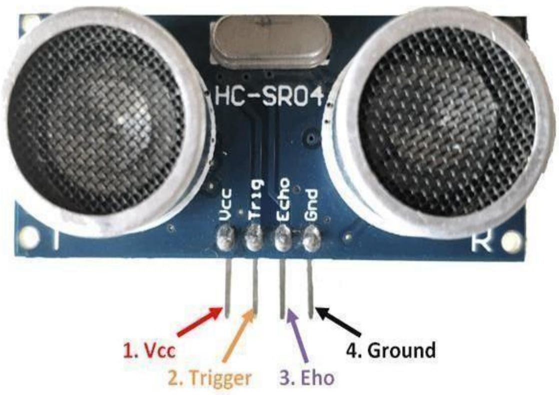


Fig 4.2.2: Ultrasonic sensor Pinout

Pin No.	**Pin Name**	**Description**
1	Vcc	The Vcc pin powers the sensor, typically with +5V
2	Trigger	Trigger pin is an Input pin. This pin has to be kept high for 10us to initialize measurement by sending US wave.
3	Echo	Echo pin is an Output pin. This pin goes high for a period of time which will be equal to the time taken for the US wave to return back to the sensor.
4	Ground	This pin is connected to the Ground of the system.

HC-SR04 Sensor

Features:

- Operating voltage: +5V
- Theoretical Measuring Distance: 2cm to 450cm
- Practical Measuring Distance: 2cm to 80cm
- Accuracy: 3mm

4.3 PINOUT AND PIN DESCRIPTION OF 16X2 LCD MODULE:

16×2 LCD is named so because; it has 16 Columns and 2 Rows. There are a lot of combinations available like, 8×1, 8×2, 10×2, 16×1, etc. But the most used one is the 16*2 LCD, All the above mentioned LCD display will have 16 Pins.

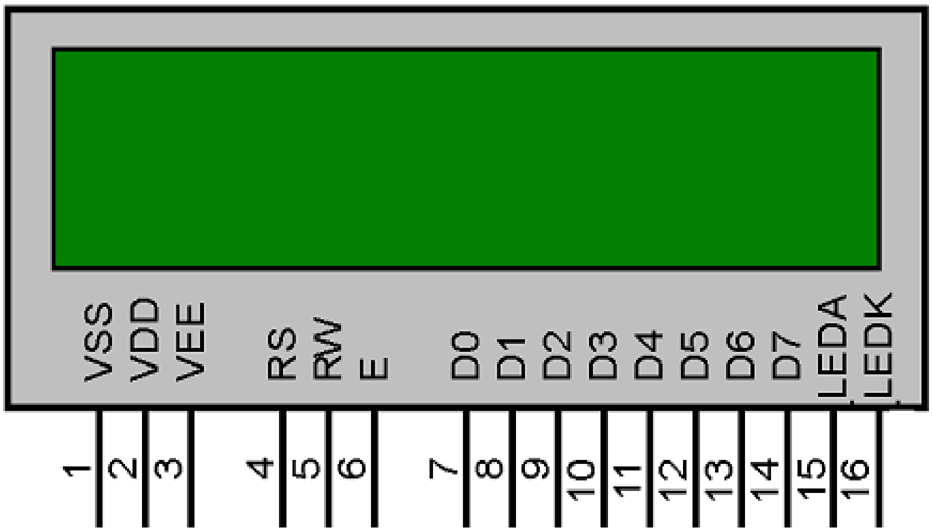


Fig 4.3.1: Pinout and Pin Description of 16x2 LCD Module

Pin No.	Pin Name	Pin Type	Pin Description	Pin Connection
Pin 1	Ground	Source Pin	This is a ground pin of LCD	Connected to the ground of the MCU/ Power source
Pin 2	VCC	Source Pin	This is the supply voltage pin of LCD	Connected to the supply pin of Power source

Pin 3	+	Control Pin	Adjusts the contrast of the LCD.	Connected to a variable POT that can source 0-5V
Pin 4	Register Select	Control Pin	Toggles between Command/Data Register	Connected to a MCU pin and gets either 0 or 1. 0 -> Command Mode 1-> Data Mode
Pin 5	Read/Write	Control Pin	Toggles the LCD between Read/Write Operation	Connected to a MCU pin and gets either 0 or 1. 0 -> Write Operation 1-> Read Operation
Pin 6	Enable	Control Pin	Must be held high to perform Read/Write Operation	Connected to MCU and always held high.
Pin 7-14	Data Bits (0-7)	Data/Command Pin	Pins used to send Command or data to the LCD.	In 4-Wire Mode Only 4 pins (0-3) is connected to MCU In 8-Wire Mode All 8 pins(0-7) are connected to MCU
Pin 15	LED Positive	LED Pin	Normal LED like operation to illuminate the LCD	Connected to +5V
Pin 16	LED Negative	LED Pin	Normal LED like operation to illuminate the LCD connected with GND.	Connected to ground

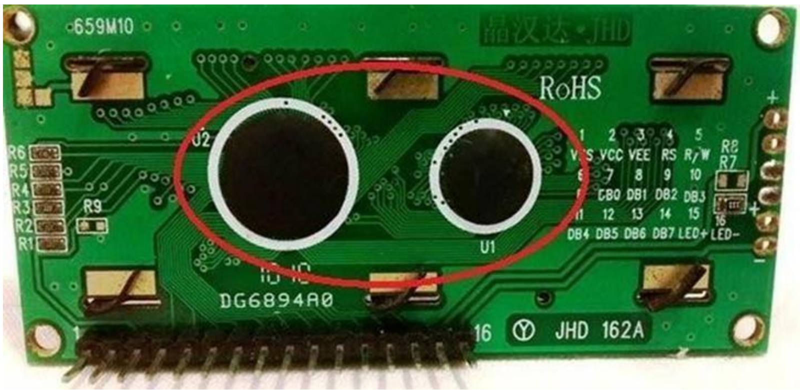


Fig 4.3.2: JHD 162A LCD

These black circles consist of an interface IC and its associated components to help us use this LCD with the MCU. Because our LCD is a 16*2 Dot matrix LCD and so it will have (16*2=32) 32 characters in total and each character will be made of 5*8 Pixel Dots.

Fig 4.3.3: Single character Pixels

It will be a hectic task to handle everything with the help of MCU, hence an **Interface IC like HD44780** is used, which is mounted on LCD Module itself. The function of this IC is to get the **Commands and Data** from the MCU and process them to display meaningful information onto our LCD Screen.

4-bit and 8-bit Mode of LCD:

The LCD can work in two different modes, namely the 4-bit mode and the 8-bit mode.In **4 bit mode** we send the data nibble by nibble, first upper nibble and then lower nibble. For those of you who don’t know what a nibble is: a nibble is a group of four bits, so the lower four bits (D0-D3) of a byte form the lower nibble while the upper four bits (D4-D7) of a byte form the higher nibble. This enables us to send 8 bit data. Whereas **in 8 bit mode** we can send the 8-bit data directly in one stroke since we use all the 8 data lines.

Read and Write Mode of LCD:

The LCD itself consists of an Interface IC. The MCU can either read or write to this interface IC. Most of the times we will be just writing to the IC, since reading will make it more complex and such scenarios are very rare. Information like position of cursor, status completion interrupts etc.

LCD Commands:

There are some preset commands instructions in LCD, which we need to send to LCD through some microcontroller. Some important command instructions are given below:

Hex Code	**Command to LCD Instruction Register**
0F	LCD ON, cursor ON
01	Clear display screen
02	Return home
04	Decrement cursor (shift cursor to left)
06	Increment cursor (shift cursor to right)
05	Shift display right
07	Shift display left
0E	Display ON, cursor blinking
80	Force cursor to beginning of first line
C0	Force cursor to beginning of second line

83	Cursor line 1 position 3
3C	Activate second line
08	Display OFF, cursor OFF
C1	Jump to second line, position 1
OC	Display ON, cursor OFF
C1	Jump to second line, position 1
C2	Jump to second line, position 2

4.4 Motor Driver Module (L298N):

The L298N is a dual H-Bridge motor driver which allows speed and direction control of two DC motors at the same time. The module can drive DC motors that have voltages between 5 and 35V, with a peak current up to 2A.

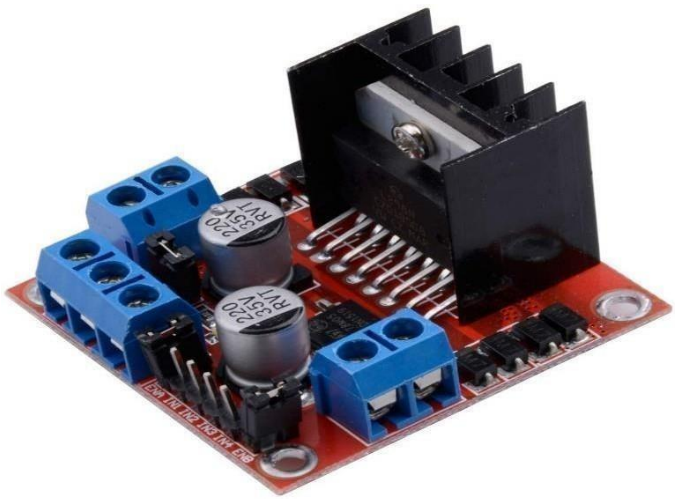


Fig 4.4.1: Motor Driver Module (L298N)

L298N Module Pinout Configuration

Pin Name	Description
IN1 & IN2	Motor A input pins. Used to control the spinning direction of Motor A
IN3 & IN4	Motor B input pins. Used to control the spinning direction of Motor B
ENA	Enables PWM signal for Motor A
ENB	Enables PWM signal for Motor B
OUT1 & OUT2	Output pins of Motor A
OUT3 & OUT4	Output pins of Motor B
12V	12V input from DC power Source
5V	Supplies power for the switching logic circuitry inside L298N IC
GND	Ground pin

Features & Specifications

- Driver Model: L298N 2A
- Driver Chip: Double H Bridge L298N
- Motor Supply Voltage (Maximum): 46V
- Motor Supply Current (Maximum): 2A
- Logic Voltage: 5V
- Driver Voltage: 5-35V
- Driver Current:2A
- Logical Current:0-36mA
- Maximum Power (W): 25W

- Current Sense for each motor
- Heatsink for better performance
- Power-On LED indicator

DC Motors

DC motor, also known as a direct current motor, is an electric motor that converts mechanical energy from the electrical energy of direct current.

Fig 4.4.2: DC Motor

IN1	IN2	Motor Behavior
		BRAKE
1		FORWARD
	1	BACKWARD
1	1	BRAKE

Servo Motor:

A servo motor (servomotor) is a highly specialized motor designed for precise control of rotary or linear motion. It's a rotational or translational motor that employs a feedback mechanism to ensure exact positioning, typically using a control signal that dictates the motor's movement to a desired position. This mechanism allows for precise control of various components, making servo motors crucial in applications where precise positioning and smooth motion are required.

Fig 4.4.3: Servo Motor

Servo Motor Pinout:

Commonly used servo motor typically have three main pins: power, ground, and signal. The power pin provides the necessary voltage to operate the motor, while the ground pin ensures a complete circuit for electricity flow. The signal pin receives control signals from an external device to determine the motor's position.

+VCC (**Red**)	Connect +VCC supply to this pin. For 9g Micro Servo it is 4.8 V (~5V).
Ground (**Brown**)	Connect Ground to this pin.
Control Signal (**Orange**)	Connect PWM of the 20ms (50 Hz) period to this pin.

Fig 4.4.4: Servo Motor Pinout

Chassis Frame*:*

A vehicle frame, also historically known as its chassis, is the main supporting structure of a motor vehicle to which all other components are attached, comparable to the skeleton of an organism.

.

Fig 4.4.5: Chassis Frame

4.5. POWER SUPPLY:

Regulated Power Supply:

Virtually every component of an electronic system converts DC electricity. Therefore, the DC power source will be necessary for each of these stages. A battery can power any system that uses little electricity. However, batteries may not be the most straightforward or affordable option for gadgets that need constant power. An unregulated power supply, which includes a filter, rectifier, and transformer, is the most effective option. Down below, you can see the diagram.

The electrical circuits inside any given device must be capable of supplying a consistent DC voltage within the device's specified power supply limit. Both the voltage and the current may only go as high as this DC supply allows. The problem is that the electrical devices are vulnerable to breakdowns caused by fluctuations in the mains supply.

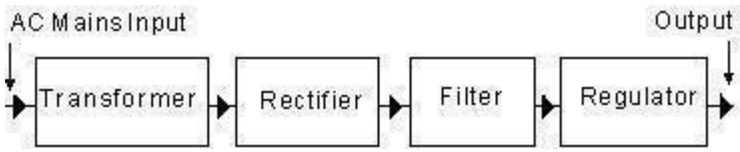


Fig 4.5.1: Block diagram of Regulated power supply

The DC power supply must reach a certain operating point, sometimes called the Q-point or quiescent point, for all electronic devices, active and passive. To lower the voltage to a level that electronics can handle, a step down transformer is used. A 230 volt 1 Ø supply is accessible in India. After passing through a rectifier, the sinusoidal AC voltage that the transformer produces becomes pulsing DC.This output is used to reduce AC ripples by sending the DC components via a filter circuit. But there are several downsides to using an uncontrolled power source.

No matter what happens to the mains alternating current (ac) or how much the load fluctuates, a regulated power supply electrical circuit will keep the voltage between the load terminals at a constant direct current (dc). A regulated power supply is an ordinarypower supply with a voltage regulator connected to it, as seen in the image. A conventional power supply supplies current to a voltage regulator, which in turn produces the desired output. An output voltage remains constant regardless of changes to either the ac input voltage or the output (or load) current.

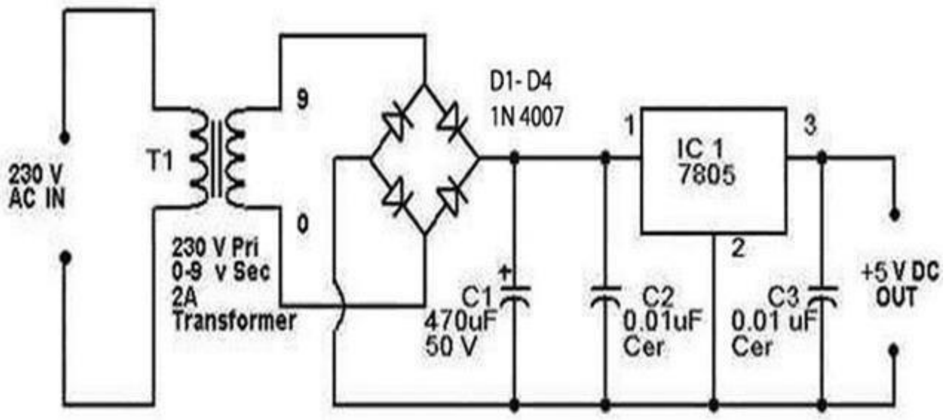


Fig 4.5.2: Circuit diagram of Regulated power supply

Wheels:

The wheels on an Autonomous Precision Farming Robot (APFR) are crucial for its mobility, stability, and efficiency in navigating complex and variable agricultural environments. They facilitate a range of tasks, from planting and weeding to monitoring crop health and harvesting


Fig 4.5.3: Wheels

4.6. SOFTWARE COMPONENTS:

Arduino IDE: For programming the microcontroller

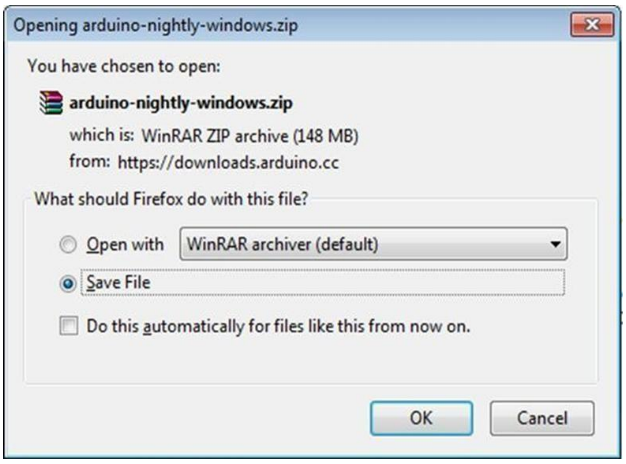
How To Upload The Arduino Board Program:

Once you've followed these easy instructions, the Arduino IDE will be running on your computer and the board will be prepared to accept software in a USB connection.

Step 1: An Arduino board—or any other suitable board—and a USB cable are the primary necessities. To link a USB printer with an Arduino Uno, a Duemilanove, Mega2560, or Diecimila, you'll need a standard USB cable...

Fig 4.6.1 USB Cable

Step 2: Arduino IDE software may be downloaded.



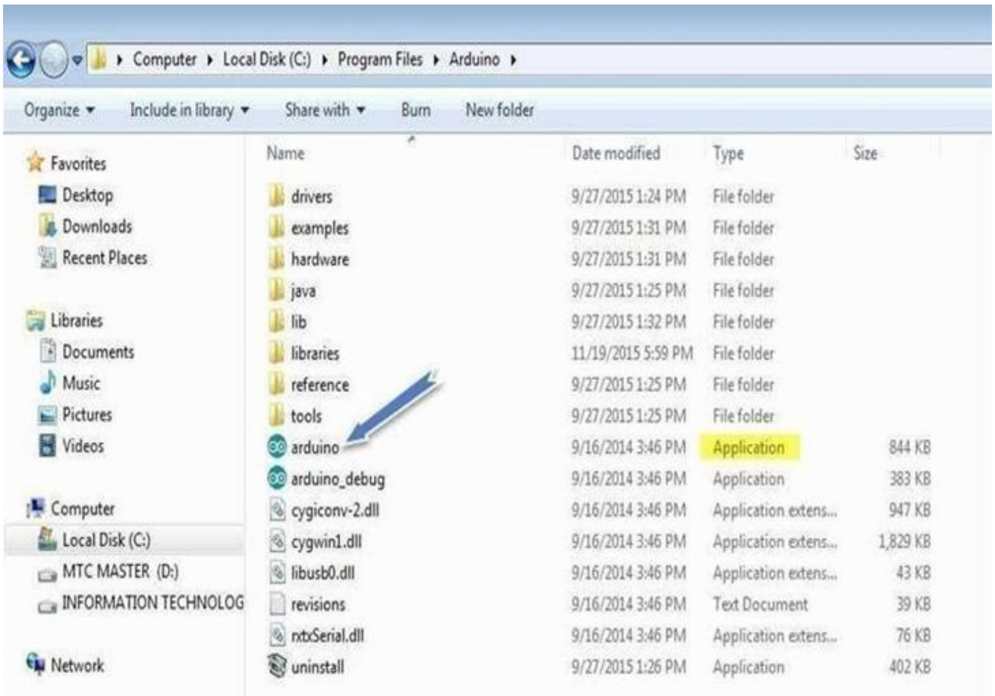
You may get many versions for the Arduino IDE on the Download page of the Arduino organization. Choose an app that is compatible with your OS, whether it iOS, Windows, or Linux. When the download is complete, open the archive.gz file.

Step 3: Activate your board.

The Micro, Duemilanove, Arduino Uno, or Mega systems have the capability to either take power form external sources or establish a connection to a computer's USB port. In order to use the Leonardo Diecimila board, it is necessary to set it to accept power via the USB cord.

Step 4: Open the Arduino Integrated Development Environment (IDE).

Upon completion of the code download, the first action is to extract the folder that encompasses the Arduino IDE. The eternally named program icon (application.exe) may be found inside this folder. To start the Integrated Development Environment (IDE), just perform a double-click action on the icon.

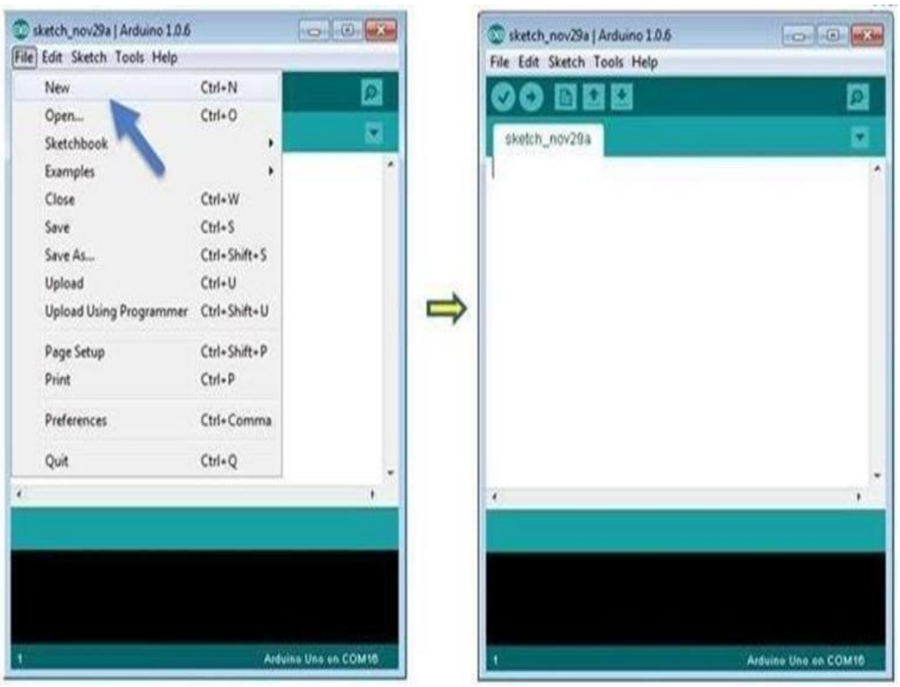


Step 5: Open your first project

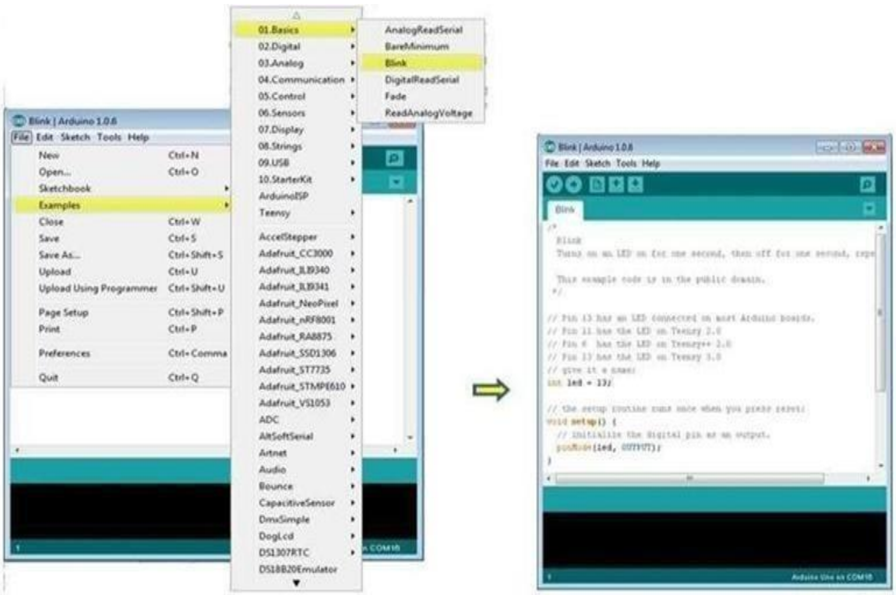
Once the software starts, you have two options : Create a new project and open an existing project

To create a new project, select File → New, To open.

There is no text. By doing this, it delays the LED's cycle of turning on and off. Instead, you might use one of the alternative instances that are accessible.

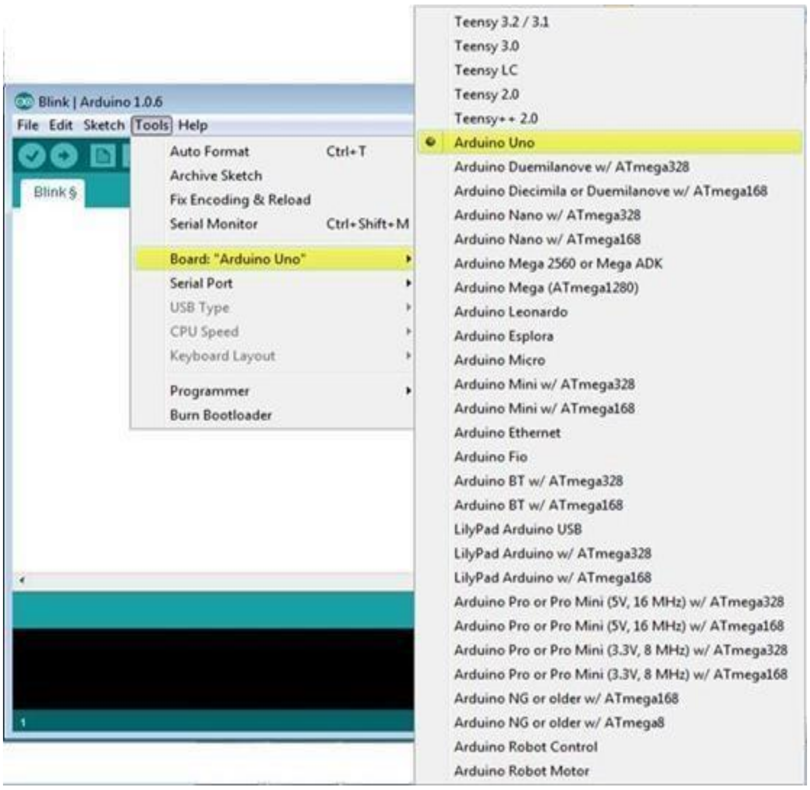


Here, we are selecting a Blink among a vast array of options. In this manner, the activation and deactivation sequence for the Led' s will be postponed. Alternatively, one might use a different of the readily available case studies.



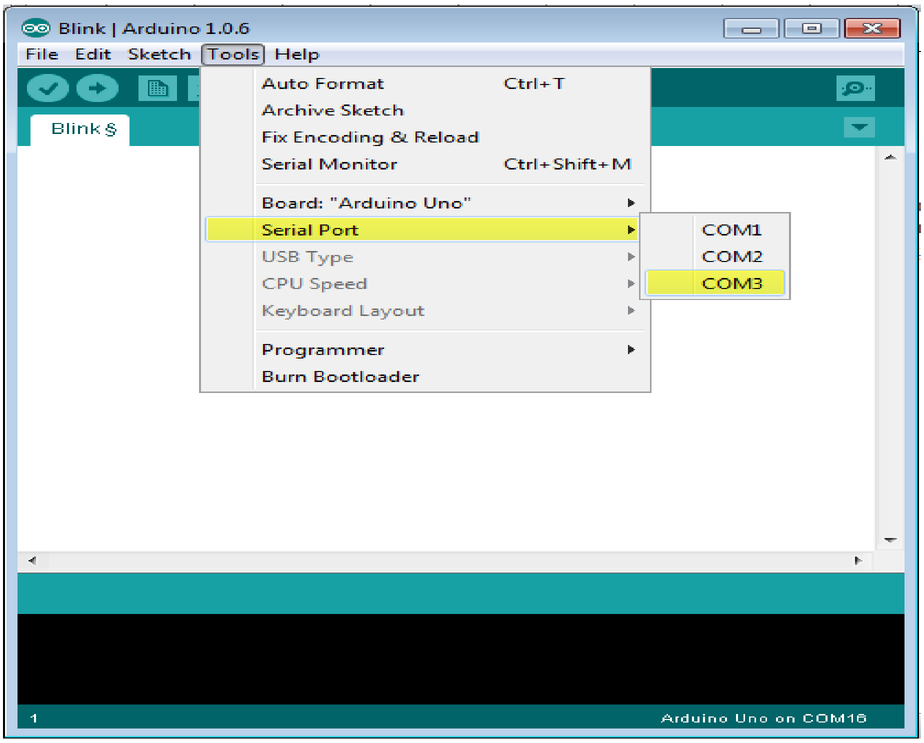
Step 6: Select your Arduino Board.

To Avoid any Error while uploading your program to the board, you must select the correct arduino board name, which matches with the board connected to the computer. Go to Tools Board and Select your Board.

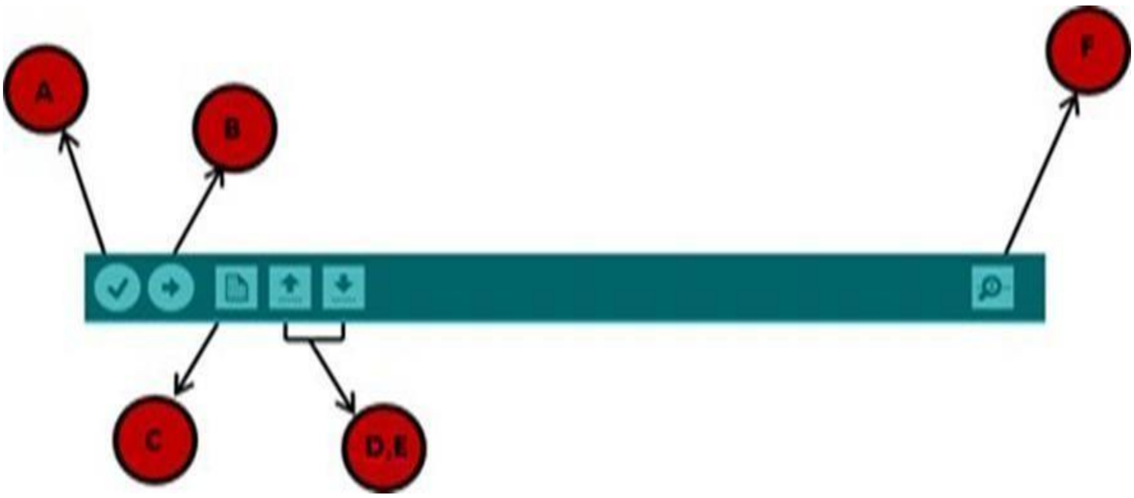


Step 7: Locate the serial connection on your device. Confirm whether a serial port has been chosen on the board that runs Arduino. To get to the "Tools" menu, choose the option for accessing your Uart Port. It is extremely probable this is COM3, because COM1 or COM2 are often used for hardware serial connections. Once you have uninstalled the Arduino board, we may dismiss the menu and check whether the corresponding item has been removed.

If you have reestablished the connection with the board, you have the liberty to choose the desired serial port.



Step 8: Submit the software.



Prior to addressing the process of uploading our software to the board, it is essential to demonstrate the functionality of each icon found on the Arduino IDE toolbar.

A-It verifies whether there were any errors throughout the compilation process. B-A software application for manipulating the Arduino microcontroller board with programming instructions.

Pressing the C key on a keyboard initiates the creation of a new drawing.To load any of the sample drawings, just double-click hereto save your drawing. time, the RX as well as Texas LEDs on the board will start to flash. After the upload process is finished, the status bar will display the message "Done uploading". The F-A serial monitor, denoted by the initial "F," is responsible for both receiving and transmitting the serial data of the board.

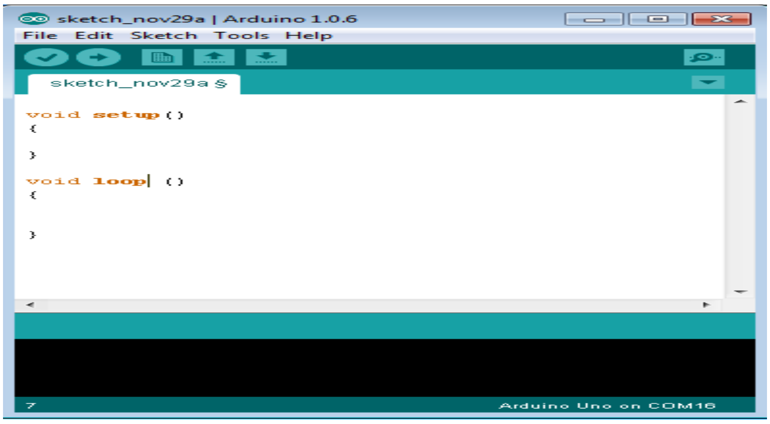
ARDUINO CODE STRUCTURE:

Learn the ins and outs of the Arduino language of programming and its associated terminology in this comprehensive chapter. Arduino is an open-source hardware platform. The C/C++ pic libraries and the Java runtime environment's source code are both made freely accessible under licences.

An Arduino programming tool is a new technological tool, and the name "sketch" characterizes it. An Arduino program primarily consists of three parts: structure, contents (constants or variables), and functions. If you want to write code for the Arduino platform that doesn't crash during compilation or have syntax errors, this is the book for you.

The Structure is where we should begin. There are primarily two purposes to software architecture:

- The function Setup
- Iterative procedure



Void setup ( )

{

}

Void Loop ( )

{

}

CHAPTER 5

EMBEDDED SYSTEMS

5.1. INTRODUCTION TO EMBEDDED SYSTEMS

The reliance on "embedded systems" in digital technology is increasing on a daily basis. Over ninety-eight percent of the world's CPUs are located in things that the average person would never think of as desktop computers. "Embedded" implies that a computer is "dedicated" to the gadget or platform it runs on. Embedded systems are purpose-built to execute a limited range of predetermined tasks, in contrast to general- purpose computers such as personal computers. Engineers may refine the mechanism for its intended use, which might reduce the product's cost and environmental effect. Companies may take advantage of advantages of scale when they mass-produce embedded systems.

The increasing usage of PC hardware is one of the many notable advances in high-end embedded technologies in the last several years. This shift has made high-end systems more accessible, allowing for initiatives that would have been impossible to fund with the exorbitant cost of non-PC-based embedded technology in the past. The PC platform has nice hardware, but there aren't any good software options.

The software in an embedded system is usually stored in read-only memory (ROM) on a single CPU board. An embedded system is used by almost all electrical devices that have a digital interface. Some examples of such devices include microwaves, watches, video recorders, and automobiles. Despite often having their own operating system, the reasoning within numerous embedded systems is so specialized that it just takes one line of code to write. Embedded systems range in size from small electronic wristwatch that sync to MP3 players to large infrastructure required to operate traffic lights, factories, and sometimes nuclear power stations. The term "firmware" is utilized on occasion to describe the computer programs installed on these CPUs.

Hardware, such as a custom-built computer circuit or an integrated circuit designed specifically for an application (ASIC). By transmitting information from various sensors and detectors to actuators, it is possible to regulate the flow of fuel to an engine or control the functioning of a machine. In an embedded system, software and hardware collaborate.

WHAT IS AN EMBEDDED SYSTEM?

We claim that a system includes embedded capability when its software is built to communicate with its hardware components throughout development. Variation and usefulness are made possible via programming.

Memory, processing units, application-specific integrated circuits, and other similar components are examples of hardware. “Embedded system" may imply whatever you want it to signify, and all of those meanings can be merged into one. Computers with a specific, limited function are known as embedded systems.

SYSTEMS THAT ARE EMBEDDED

Embedded systems have many different kinds of uses. The once-pricing industrial control applications, as well as the automotive, communication, and office appliance industries, as well as the television industry, have recently begun to employ specialist processors.

1. The veryearliest teller machines
2. Talking over the phone and using cellular networks
3. Hardware components and network interface cards
4. Printers for computers
5. Hard discs
6. Items for automating the house
7. Compact mathematical devices
8. Household appliances
9. instruments used in medicine
10. Measurement instruments
11. Watches that can do many tasks
12. Multi-Use Printers

5.2 EMBEDED SYSTEMS HISTORY

Among the first known examples of an orbital guidance computer, created by “Charlie Stark Draper" of the Massachusetts Institute of Technology Measurement Laboratory. The most important component of the project, the Apollo guidance computer, used integrated circuits and newly developed monolithic circuits to decrease size and weight. One of the earliest mass- produced embedded devices was the Autonetics the defendant-17 navigation processor, which was used in the 1961 Minuteman missile launch. Its main memory was kept on a hard drive, and it was built using transistor logic. In 1966, at the same time as construction of the Minuteman II was underway, an upgraded computer that heavily used integrated circuits supplanted the D-17. This effort was instrumental in reducing the price of quad & gate semiconductors from $1,000 to $3, which allowed them to be included in mass-produced devices.

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5.3 EMBEDDED SYSTEMS FEATURES

Embedded computer systems provide a high level of adaptability, allowing them to streamline many processes ranging from product creation to manufacturing. Consequently, they have the potential to be beneficial in any company setting. Specialized operating systems in embedded devices are crucial for intricate, functionality-packed systems. An embedded operating system is a lightweight system that can replicate the functionality of a real-time operating system.

There are instances when embedded systems lack the same level of power or complexity as general-purpose systems. In embedded systems, it is customary to use a CPU with reduced RAM capacity and low power consumption. Embedded devices often use compact operating systems, despite their limited capability.

Design engineers may enhance the performance of an embedded system by minimizing its physical size while simultaneously improving its effectiveness, speed, and longevity, as it is used only for its designated purpose. A widely adopted approach is to manufacture sure embedded systems in large quantities to take advantage of economies of scale. Regrettably, certain embedded systems are immune to environmental hazards. High humidity and extreme heat are only two instances. Quiet operation is advantageous for devices that possess substantial computing power, such as mobile phones and mobile music players.

Cost is a significant factor to consider in design. Typically, engineers choose for hardware that meets the required capabilities adequately. By implementing an operating system that is real-time or restricting the available programs on general-purpose hardware, it becomes possible for low-volume and trial integrated systems to operate.

5.4 EMBEDDED SYSTEM CHARACTERISTICS

Central processing units (CPUs) are often built into embedded computing systems because of the complex features they generally include. Their many non-functional requirements, however, make the work much more difficult.

- Timely completion is essential to avoid system failure
- Multiple rates of operation
- Minimal power usage in a number of cases
- Reducing code size leads to cheaper manufacture.

Functions are often given precedence by programmers focusing on workstations. Even while they may test out sure processing kernels inside their application, they hardly never test out the product's overall performance. The production cost and power use are almost never taken into account.

5.5 A SYSTEM OF INTEGRATED COMPONENTS

In any system with embedded functionality, you may find a CPU (central processing unit) and other purpose-built hardware. The memory chips that hold the code are part of this hardware. The term "firmware" may refer to either the software or the actual contents of memory chips. An embedded system's architecture may be conceptualized using a layered framework, displayed in Figure 19.

Any device, such as a desktop machine, laptop, or tablet, may use this layout. The difference between the two must be clearly defined, however. Every embedded system doesn't need its own operating system. Toys, cooling systems, remote controls, and other tiny devices don't need an operating system when their software is developed.

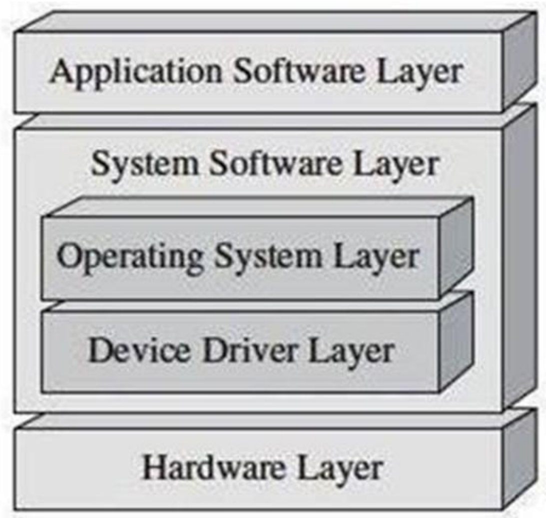


Fig 5.5.1: Embedded System Layered Architecture

It is recommended to use an operating system while running applications that need sophisticated processing. Inserting the program into the memory chip follows the integration with the operating system. It is not possible to remove software from the memory of a machine chip after it has been installed. Let's take a look at an embedded system's hardware and see how it works. There is no deviation from the figure at all. Core members of the team. Devices for input and output, memory (both random access and read-only), a means of communication, and circuits tailored to certain applications.

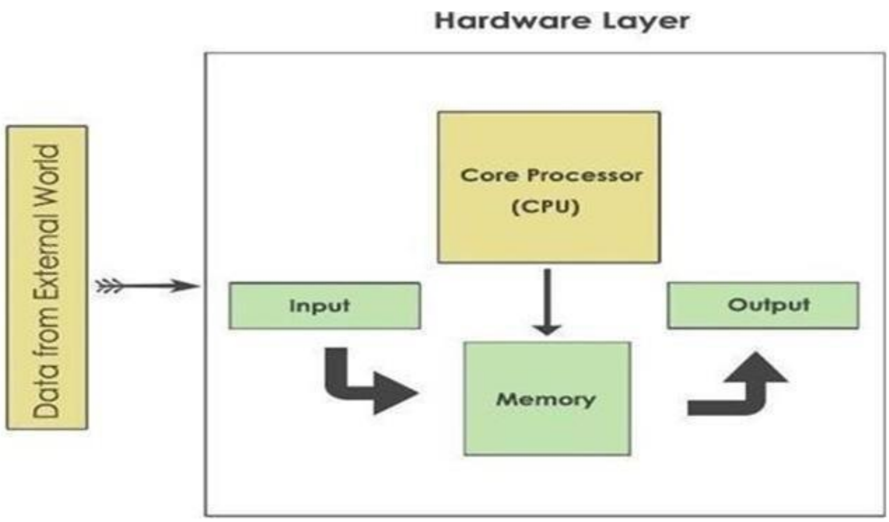


Fig 5.5.2: Diagram of Embedded systems

5.6 A CPU, OR CENTRAL PROCESSING UNIT,

A variety of choices are available, including processing of digital signals, microcontrollers, and microprocessors. The DSP serves as the brains of the operation, processing digital signals. Micro controllers are small and inexpensive computer chips. One of the main selling points is that a single chip will include memory, a digital to analog converter, a series of communications interface, and a plethora of other features.

It is appropriate for tiny applications to use a micro-controller since it does not need many other components. However, microprocessors cannot function without a plethora of auxiliary components. Many signal processing applications rely on D5P, including those dealing with audio and video.

Memory:

The two most common types of computer memory are RAM (random access memory) and read-only memory (ROM). If a device loses power, all data stored in memory will be wiped. However, data that can only be read will remain untouched. The result is that the ROM has the firmware. The program is loaded into the central processor unit's (CPU) read-only memory (ROM) when power is applied.

Devices for entering data:

When compared to desktop computers, embedded systems' input devices are much weaker. It is not easy to communicate with the embedded system of the computer since there is no physical keyboard and mouse. Embedded electronics often take the form of a tiny keypad with a handful of buttons.

Devices for producing results:

The output devices of embedded systems also have limited functionality. To display the status of individual components' operations or to indicate the existence of an alarm, embedded systems may use light-emitting diodes (LEDs). Alternatively, a small LCD (liquid crystal display) might indicate certain parameters.

Interfaces for communication:

It is not impossible for embedded devices to communicate with one another or for data to be sent to a desktop computer. One way embedded systems make this possible is by including one or more communication interfaces, including USB, IEEE1394, Ethernet, RS422, RS485, or RS232.

Dedicated Circuitry For A Particular Application:

Embedded systems may be equipped with transducers, sensors, and specialized processing and oversight circuits to tackle various tasks. The only way that this circuit and the CPU to communicate is for the CPU to do the essential tasks. All of the machinery needs either a 230- volt outlet or a battery pack to keep running. When making hardware, it's best to aim for the lowest possible power usage.

Applications

- Smart Irrigation Systems
- Water Tank Management
- Greenhouses
- Industrial Cooling Systems
- Crop Row Navigation
- Obstacle Detection
- Smart Monitoring
- Precision Seeding or Spraying
- Educational Tool
- Small-Scale Farming

CHAPTER 6

6.1. SOURCE CODE

```
#include <Servo.h>

const int trigPinFront = 9;
const int echoPinFront = 8;

const int trigPinSide = 12;
const int echoPinSide = 11;

const int ENA = 5;
const int IN1 = 2;
const int IN2 = 3;
const int ENB = 6;
const int IN3 = 4;
const int IN4 = 7;

Servo taskServo;
const int servoPin = 10;

int frontDistance = 0;
int sideDistance = 0;
const int OBSTACLE_DISTANCE = 20; // cm

void setup()
{

  Serial.begin(9600);
  Serial.println("Precision Farming Robot Initialized");

  pinMode(trigPinFront, OUTPUT);
  pinMode(echoPinFront, INPUT);
  pinMode(trigPinSide, OUTPUT);
```

```
  pinMode(echoPinSide, INPUT);
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  taskServo.attach(servoPin);
  taskServo.write(90);

  delay(1000);
}

void loop() {
  frontDistance = readUltrasonic(trigPinFront, echoPinFront);
  sideDistance = readUltrasonic(trigPinSide, echoPinSide);



  Serial.print("Front: ");
  Serial.print(frontDistance);
  Serial.print(" cm\tSide: ");
  Serial.print(sideDistance);
  Serial.println(" cm");

  if (frontDistance < OBSTACLE_DISTANCE)
  {
    Serial.println("Obstacle Detected! Turning Right...");
    stopMotors();
    delay(500);
    turnRight();
    delay(700);
    stopMotors();
  }
```

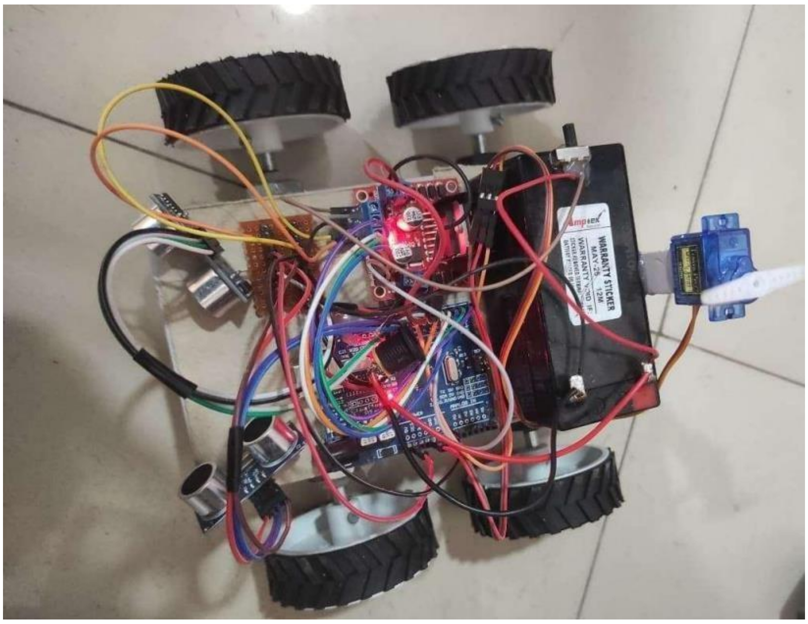
```
  else if (sideDistance > 25)
  {
    Serial.println("Too far from row - Turning Left");
    turnLeft();
    delay(300);
    moveForward();
  }
  else if (sideDistance < 10)
  {
    Serial.println("Too close to row - Turning Right");
    turnRight();
    delay(300);
    moveForward();
  }
  else
  {
    Serial.println("Path Clear - Moving Forward");
    moveForward();
  }

  Serial.println("Activating Servo - Spraying...");
  taskServo.write(0);
  delay(300);
  taskServo.write(90);
  delay(500);
}


int readUltrasonic(int trigPin, int echoPin)
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
```

```
digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;
  return distance;
}
void moveForward()
{
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);

  analogWrite(ENA, 150);
  analogWrite(ENB, 150);
}

void stopMotors()
{
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);

  analogWrite(ENA, 0);
  analogWrite(ENB, 0);
}

void turnLeft()
{
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
```

```
digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);

  analogWrite(ENA, 150);
  analogWrite(ENB, 150);
}

void turnRight()
{
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);

  analogWrite(ENA, 150);
  analogWrite(ENB, 150);
}
```

6.2 RESULT

- Successful detection of obstacles in the field.
- Accurate navigation along crop rows.
- Continuous real-time monitoring of field conditions.
- Reduced human intervention in farming operations.



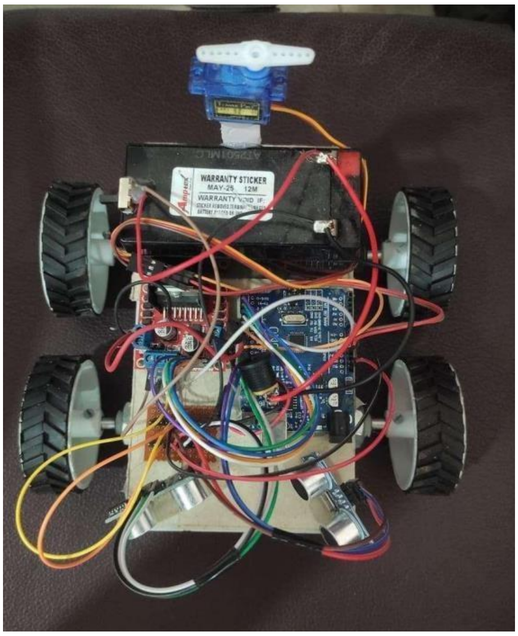


Fig 6.2.1 Autonomous Robot for smart crop management

CHAPTER 7

CONCLUSION

This project demonstrates how an Arduino-based autonomous robot can meaningfully contribute to the goals of precision agriculture by integrating navigation, sensing, and actuation in a cohesive platform. By leveraging ultrasonic sensors for obstacle detection, DC and servo motors for mobility and alignment, and a 16×2 LCD for real-time feedback, the system enables automated field operations such as crop-row alignment and environmental monitoring. The real-time data acquisition and decision-making capabilities built into the robot support more efficient use of inputs, reduced reliance on labour, and improved crop productivity. Moreover, thanks to its modular architecture, the robot can be adapted to a variety of crops and varied terrain conditions, making it a flexible tool for modern smart-farming applications. Ultimately, the work provides a scalable prototype that aligns with broader aims of sustainable agriculture: increasing output, lowering input and labour costs, and enhancing resource-use efficiency.

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